Flight Delay & Cancellation Analysis Project

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Problem Statement and Objective

Problem Statement:

Airlines were facing a surge in customer complaints due to frequent flight delays and cancellations. Leadership lacked visibility into the root causes, with unclear metrics caused by missing or inconsistent data in departure times and delay records. Without structured analysis, it was difficult to identify whether delays were concentrated at specific airports, routes, or times of day, and cancellation trends by airline remained hidden. This created operational inefficiencies and negatively impacted customer satisfaction.

Objective:

- Clean and validate flight data using SQL to ensure accuracy in delay and cancellation metrics.
- Analyze patterns and root causes of delays and cancellations by airline, airport, route, and time of day.
- Build interactive Power BI dashboards with KPIs and matrices to provide actionable insights, reduce complaints, and improve operational performance.

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Dataset Overview

- **Sources** :CSV
- **Size:** (500, 9)
- **Timeframe:** 2024-01-01 to 2024-31-06
- **Tools** : PostgreSQL and Power BI
- **Key Variables**:

flight_id : flight record ID

airline : Airline name

flight_number: Flight number

origin :Origin airport

destination : Destination airport

scheduled_departure : Scheduled departure time

actual_departure : Actual departure time

delay_minutes : Delay in minutes

canceled : Whether flight was canceled

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Dataset

Sample Schema (Table: flights)

Column	Type	Description
flight_id	INT	Unique flight record ID
airline	TEXT	Airline name
flight_number	TEXT	Flight number
origin	TEXT	Origin airport
destination	TEXT	Destination airport
scheduled_departure	TIMESTAMP	Scheduled departure time
actual_departure	TIMESTAMP (NULL)	Actual departure time
delay_minutes	INT (NULL)	Delay in minutes
canceled	BOOLEAN (0 or 1)	Whether flight was canceled

```sql
CREATE TABLE flights (
    flight_id           INT PRIMARY KEY,
    airline             VARCHAR(50),
    flight_number       TEXT,
    origin              VARCHAR(10),
    destination         VARCHAR(10),
    scheduled_departure TIMESTAMP,
    actual_departure    TIMESTAMP NULL,
    delay_minutes       NUMERIC NULL,
    canceled            BOOLEAN
);
```

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```sql
--1.check if we have missing data in departure times or delays?
select * from flights
where actual_departure is null or delay_minutes is null;
```

Data Output Messages Notifications

Showing rows: 1 to 59 Page No: 1

	flight_id [PK] integer	airline character varying (50)	flight_number text	origin character varying (10)	destination character varying (10)	scheduled_departure timestamp without time zone	actual_departure timestamp without time zone	delay_minutes numeric	canceled boolean
1	6	Delta	7627	MIA	LAX	2024-06-26 03:36:00	[null]	[null]	true
2	12	Southwest	4486	LAX	ATL	2024-06-15 05:33:00	[null]	[null]	true
3	13	United	3583	SEA	SFO	2024-06-10 01:14:00	[null]	[null]	true
4	47	United	8766	JFK	MIA	2024-02-24 12:00:00	[null]	[null]	true
5	50	United	8380	SEA	DFW	2024-06-13 19:28:00	[null]	[null]	true
6	57	Southwest	506	ORD	DEN	2024-05-22 10:17:00	[null]	[null]	true
7	61	United	5853	MIA	SFO	2024-05-03 16:51:00	[null]	[null]	true
8	62	Alaska Airlines	5525	JFK	LAX	2024-06-08 15:50:00	[null]	[null]	true
9	65	United	827	SFO	JFK	2024-04-04 15:33:00	[null]	[null]	true
10	66	Alaska Airlines	3328	DEN	SFO	2024-01-13 17:43:00	[null]	[null]	true
11	75	Delta	7065	ATL	ORD	2024-05-26 09:12:00	[null]	[null]	true
12	76	Alaska Airlines	7711	JFK	MIA	2024-02-15 08:45:00	[null]	[null]	true
13	84	Delta	9532	SFO	SEA	2024-01-02 01:45:00	[null]	[null]	true
14	96	American Airlines	1769	DFW	LAX	2024-05-24 22:46:00	[null]	[null]	true
15	118	United	9416	PHX	JFK	2024-04-20 23:31:00	[null]	[null]	true

- Use of IS NULL, WHERE, basic filtering.
- The query identified flights with missing actual_departure or delay_minutes values. All such records were canceled, highlighting data gaps that could distort delay metrics and needed cleaning before analysis.

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```sql
--2.Calculate Average Delay by Airline
SELECT airline, ROUND(AVG(delay_minutes), 2) AS avg_delay
FROM flights
WHERE canceled = false
GROUP BY airline
ORDER BY avg_delay DESC;
```

Data Output Messages Notifications

	airline character varying (50)	avg_delay numeric
1	Alaska Airlines	48.06
2	United	46.04
3	Delta	44.53
4	Southwest	40.18
5	American Airlines	39.48

- Use of AVG(), GROUP BY, filtering with WHERE
- The query calculated average delays for non-canceled flights. Alaska Airlines showed the highest average delay (48 mins), while American Airlines had the lowest (39 mins), revealing performance differences across carriers.

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```sql
--3.Which airline has the worst average delay? Focus only on flights that weren't canceled.
SELECT
  CASE
    WHEN EXTRACT(HOUR FROM scheduled_departure) BETWEEN 5 AND 11 THEN 'Morning'
    WHEN EXTRACT(HOUR FROM scheduled_departure) BETWEEN 12 AND 17 THEN 'Afternoon'
    ELSE 'Evening'
  END AS time_of_day,
  ROUND(AVG(delay_minutes), 2) AS avg_delay
FROM flights
WHERE canceled = false
GROUP BY time_of_day
ORDER BY avg_delay DESC;
```

Data Output Messages Notifications

	time_of_day text	avg_delay numeric
1	Evening	44.84
2	Morning	43.59
3	Afternoon	42.15

- Use of EXTRACT(HOUR), CASE WHEN, grouping
- The query grouped delays by morning, afternoon, and evening departures. Results showed evening flights had the worst delays (44.8 mins), followed by morning and afternoon, indicating time-based operational bottlenecks.

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```sql
--4.Identify Most Delayed Airports or Are delays worse in the morning or evening?
SELECT origin, ROUND(AVG(delay_minutes), 2) AS avg_delay
FROM flights
WHERE canceled = false
GROUP BY origin
ORDER BY avg_delay DESC;
```

Data Output Messages Notifications

	origin character varying (10)	avg_delay numeric
1	MIA	48.86
2	PHX	48.58
3	SFO	47.37
4	DFW	46.34
5	SEA	45.24
6	DEN	44.60
7	LAX	42.37
8	ORD	41.76
9	JFK	37.46
10	ATL	34.55

The query ranked airports by average delay times. Miami (MIA) and Phoenix (PHX) showed the highest delays (~49 mins), while Atlanta (ATL) had the lowest (~35 mins), highlighting airport-specific performance issues.

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```sql
--5.Calculate cancellation rate or which airlines cancel the most flights. What are the cancellation rates?
SELECT
    airline,
    COUNT(*) AS total_flights,
    SUM(CASE WHEN canceled = TRUE THEN 1 ELSE 0 END) AS canceled_flights,
    SUM(CASE WHEN canceled = FALSE THEN 1 ELSE 0 END) AS operated_flights,
    ROUND(
        (SUM(CASE WHEN canceled = TRUE THEN 1 ELSE 0 END) * 100.0) / COUNT(*),
        2
    ) AS cancellation_rate_percent
FROM
    flights
GROUP BY
    airline
ORDER BY
    cancellation_rate_percent DESC;
```

Data Output Messages Notifications

	airline character varying (50)	total_flights bigint	canceled_flights bigint	operated_flights bigint	cancellation_rate_percent numeric
1	United	110	18	92	16.36
2	Southwest	100	13	87	13.00
3	Alaska Airlines	106	12	94	11.32
4	American Airlines	95	10	85	10.53
5	Delta	89	6	83	6.74

The query calculated cancellation rates per airline. United had the highest cancellation rate (16.36%), while Delta had the lowest (6.74%), showing clear differences in reliability across carriers.

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```sql
--6. Summary
SELECT
    airline,
    ROUND(AVG(delay_minutes), 2) AS avg_delay,
    SUM(CASE WHEN canceled = TRUE THEN 1 ELSE 0 END) AS total_cancelled,
    COUNT(*) AS total_flights,
    ROUND(SUM(CASE WHEN canceled = TRUE THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS cancel_rate_percent
FROM flights
GROUP BY airline
ORDER BY avg_delay DESC;
```

Data Output Messages Notifications

	airline character varying (50)	avg_delay numeric	total_cancelled bigint	total_flights bigint	cancel_rate_percent numeric
1	Alaska Airlines	48.06	12	106	11.32
2	United	46.04	18	110	16.36
3	Delta	44.53	6	89	6.74
4	Southwest	40.18	13	100	13.00
5	American Airlines	39.48	10	95	10.53

The summary combined average delays and cancellation rates. Alaska Airlines had the worst delays (48 mins), while United had both high delays and the highest cancellation rate, making them key focus areas for improvement.

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Flight Delay & Cancellation Analysis Dashboard

ORIGIN: All

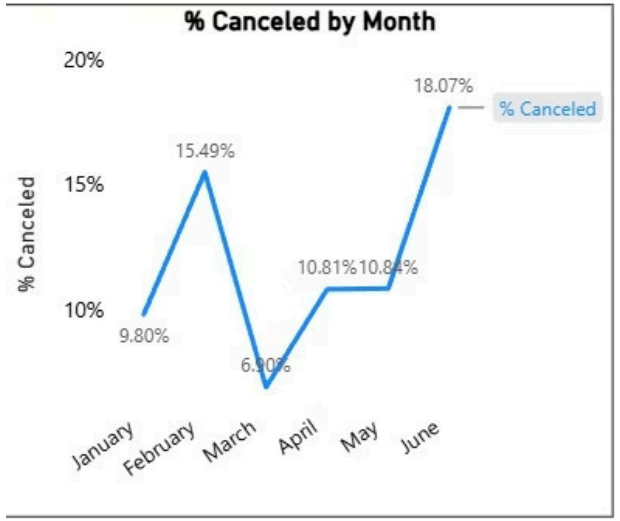
DESTINATION: All

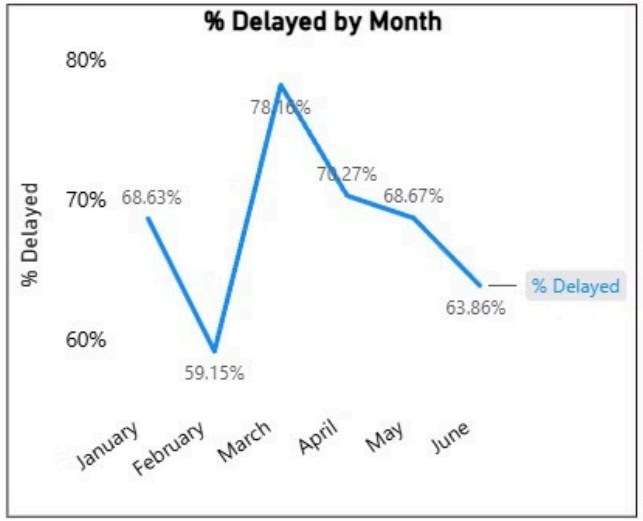
AIRLINE
- Alaska Airlines
- American Airlines
- Delta
- Southwest
- United

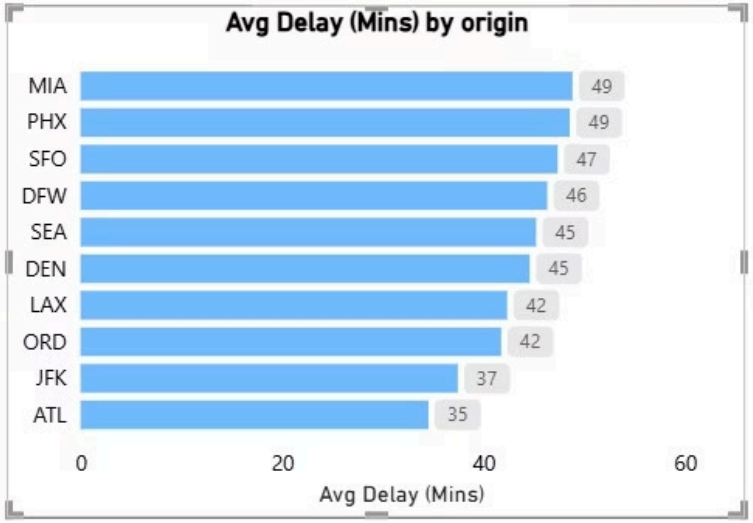
Time
- Afternoon (12 PM–6 PM)
- Morning (6 AM–12 PM)
- Evening (6 PM–12 AM)
- Night (12 AM–6 AM)

Days
- Monday
- Tuesday
- Wednesday
- Thursday
- Friday

500	11.80%	43.77	68.40%
Total Flights	% Canceled	Avg Delay (Mins)	% Delayed







Route	% Canceled	% Delayed
ATL → DEN	16.67%	50.00%
ATL → DFW	12.50%	50.00%
ATL → JFK		100.00%
ATL → LAX		80.00%
ATL → MIA	25.00%	50.00%
ATL → ORD	40.00%	40.00%
ATL → PHX		60.00%

Time	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Monday	7.50	63.25	35.75	69.00	3.00	45.33	37.50	44.67	55.50	31.33	110.00	0.00	22.67	53.00	30.00	38.83	28.50	33.88	55.00	103.00	60.50	34.67	56.00
Tuesday	29.40	41.00	54.80	42.00	48.17	47.00	61.00	31.33	40.00	36.00	33.00	32.25	118.00		20.00	14.00	52.33	35.60	84.00	8.50		63.00	14.67
Wednesday	22.00	24.00			15.00	81.00	20.33	50.00	49.20	67.83	51.50	0.00	42.00	45.25	4.00	56.33	36.86	46.75	36.33	44.00	21.00	56.67	24.67
Thursday		48.00	29.67	38.00	15.50	15.67	39.00	15.50		29.80	51.00	16.00	30.50	24.60	24.67	17.50	90.00	39.75	56.67	59.25	30.80		40.00
Friday		13.00	53.00	30.33	42.00	72.75	57.00	44.50	22.75	52.33	64.00	18.33	19.00	57.00	16.00	52.25	34.50	45.25		32.00	26.00	66.00	68.83
Saturday	38.33		40.00	52.00	28.50	18.00	37.00		30.00	86.80		71.50	14.00	39.00		91.50	89.67	31.33	11.50	103.00	3.00	36.60	44.00
Sunday	42.50	38.00	42.67	46.00	49.00	85.00	6.00	52.50	46.33	18.40		51.00	95.00	92.00	18.00	53.50	35.00	47.00		84.00	52.60	66.75	98.00

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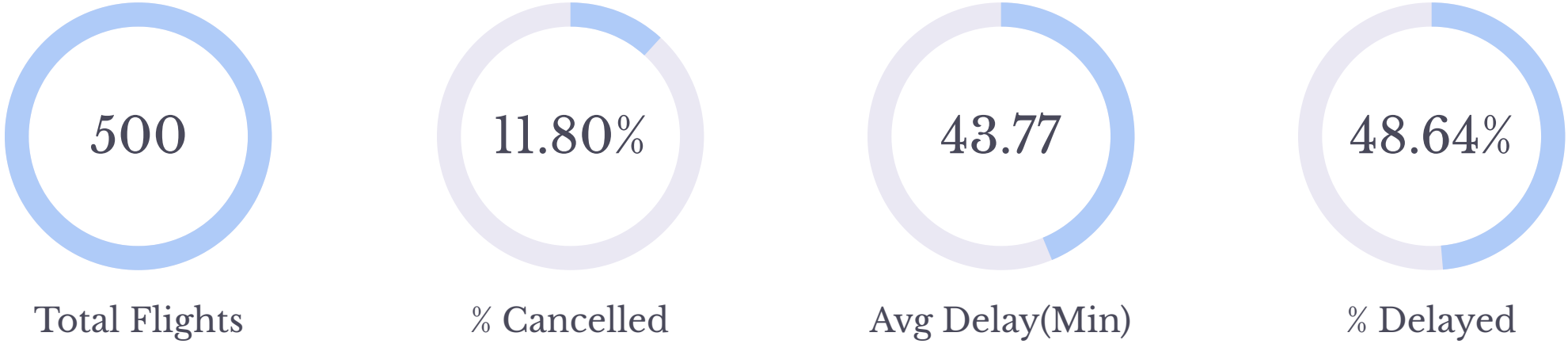
Dashboard Highlights

- Built an automated **Power BI dashboard** for Flight Delay & Cancellation Analysis with **4 KPIs**:



- Line chart shows **% Cancelled by Month** → June recorded the highest cancellation rate.
- Line chart shows **% Delayed by Month** → March had the highest delay percentage.
- Bar chart highlights **MIA airport** with the highest average delay (49 minutes).
- Table matrix displays **Route-wise % Cancelled and % Delayed**, surfacing problem routes (e.g., ATL → JFK).
- Heatmap matrix shows **Avg Delay by Hour (0–23) and Day of Week**, revealing spikes between 3–6 PM.
- Interactive slicers added for **Airline, Origin, Destination, Time, and Day filters** to drill down insights.

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Functional Uses (DAX formula)

Calculated Columns:

- *Departure Hour = HOUR('flights'[scheduled_departure])*
- *Day of Week = FORMAT('flights'[scheduled_departure], "dddd")*
- *Route = 'flights'[origin] & " → " & 'flights'[destination]*
- *Is Delayed = IF('flights'[delay_minutes] > 15, "Yes", "No")*

Key Measures:

- *Total Flights = COUNTROWS('flights')*
- *Avg Delay (Minutes) = AVERAGE('flights'[delay_minutes])*
- *Total Cancellations = CALCULATE([Total Flights], 'flights'[canceled] = 1)*
- *% Canceled = DIVIDE([Total Cancellations], [Total Flights])*
- *Delayed Flights = CALCULATE([Total Flights], 'flights'[Is_Delayed] = "Yes")*
- *% Delayed = DIVIDE([Delayed Flights], [Total Flights])*

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Executive Summary

- **Purpose:** To identify root causes of flight delays and cancellations and provide actionable insights.
- **Scope:** Analysis of 500 flights across multiple airlines, routes, and time periods.
- **Key KPIs : Total Flights:** 500

  % Canceled: 11.8%

  Avg Delay: 43.8 minutes

  % Delayed: 68.4%

Highlights:

- June recorded the highest cancellation rate.
- March showed the highest delay percentage.
- Miami (MIA) had the longest average delays.
- Evening flights (3–6 PM) experienced the worst delays.
- Route ATL → JFK had the highest cancellation rate.

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Recommendations

- **Focus on Evening Flights (3–6 PM):** Address operational bottlenecks during peak delay hours with better scheduling and resource allocation.
- **Improve Route Reliability:** Prioritize corrective measures for high-risk routes such as **ATL → JFK**, which showed the highest cancellation rate.
- **Airport-Specific Action Plans:** Target airports with consistently high delays (MIA, PHX, SFO) by reviewing ground operations and turnaround processes.
- **Airline Performance Monitoring:** Engage with carriers like **United**, which had the highest cancellation rate, to improve reliability and customer satisfaction.
- **Continuous Dashboard Use:** Leverage the Power BI dashboard for ongoing monitoring, enabling leadership to track KPIs and act on emerging trends in real time.

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